

Executive overview

30,000 Taiwanese credit card customers, Apr-Sep 2005. How big is the book, is it getting riskier, and where does the value come from? NT\$ figures marked est. are estimates.

Segment

All

Credit limit tier

All

30K

Customers

28K

Active in Sep

NT\$1.5bn

Balance (Sep)

NT\$22.0M

Revenue / mo (est.)

NT\$10.2M

Exp. loss / mo (est.)

NT\$7.4M

Risk-adj. profit / mo...

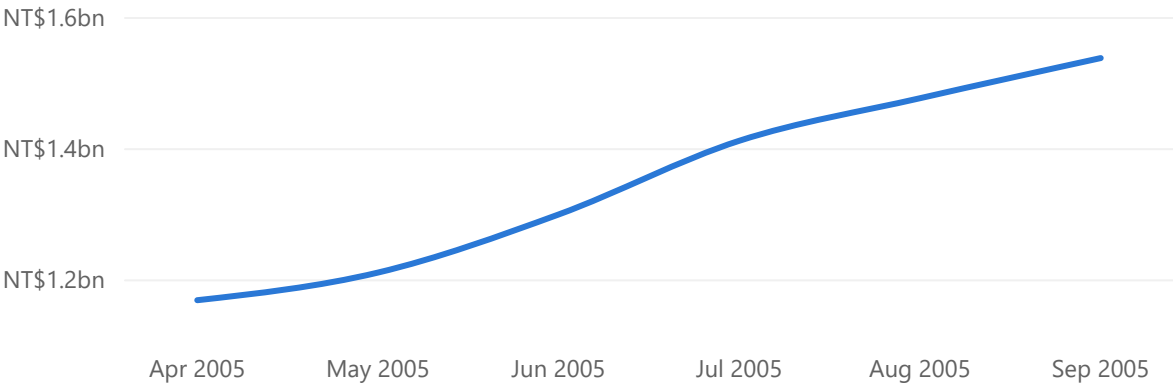
22.1%

Default rate (Oct)

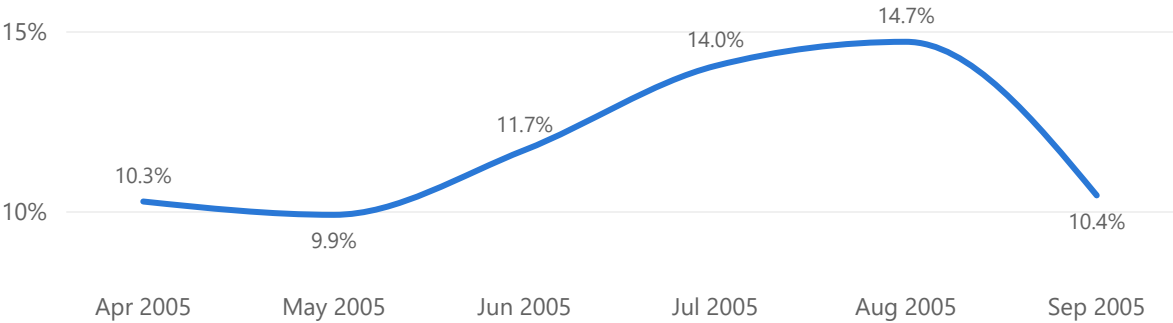
38.1%

Loss-making (est.)

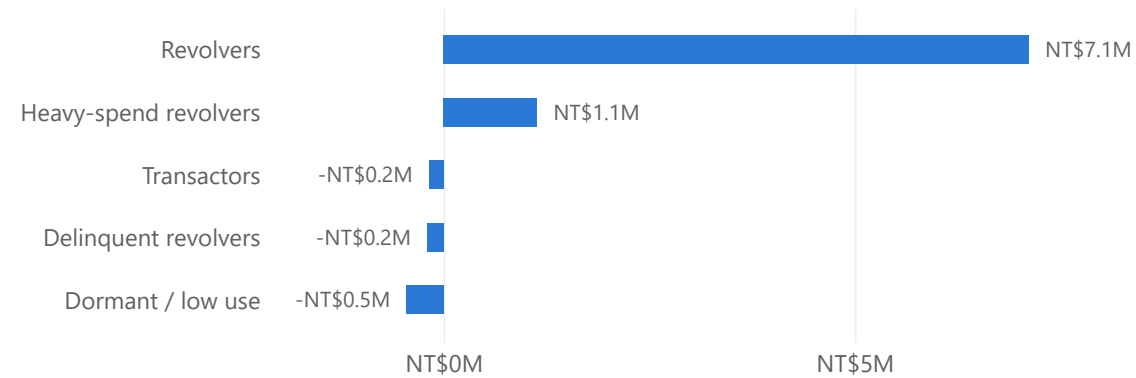
Is the book growing? Balance outstanding by month



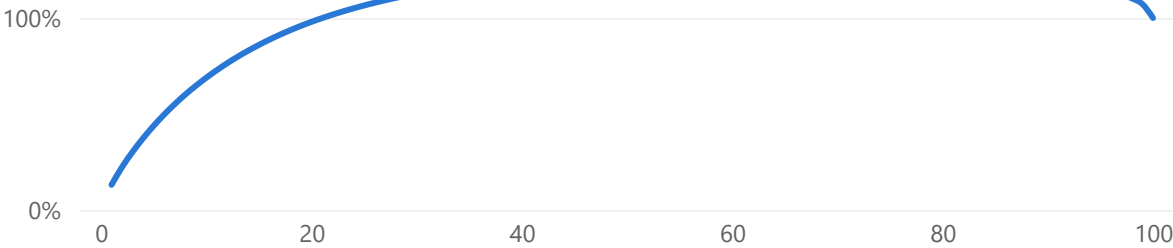
Is credit quality getting worse? Accounts 60+ days past due (Sep status codes not comparable)



Where does the value come from? Est. monthly risk-adjusted contribution by segment



How concentrated is value? Cumulative share of est. contribution from the top X% of customers (whole portfolio)



Customer analytics

Five behavioral segments from K-means on utilization, payment, activity and delinquency ratios. Who are our customers and how do they use the card?

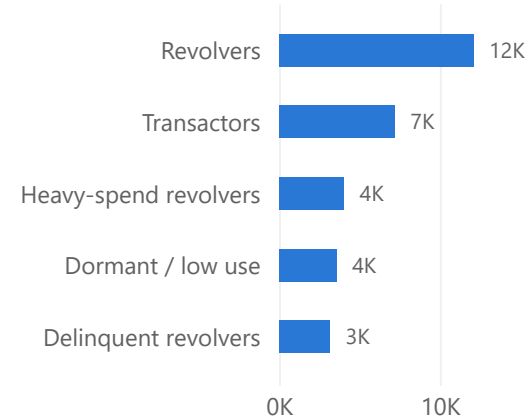
Credit limit tier

All

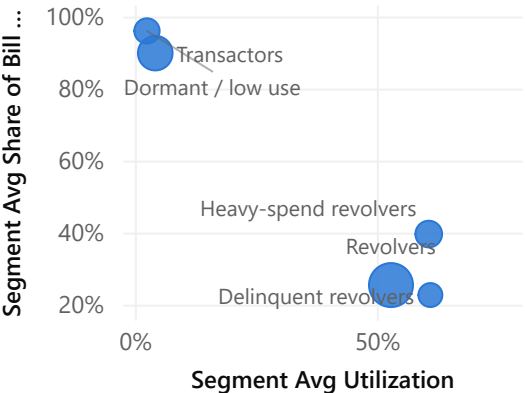
Age band

All

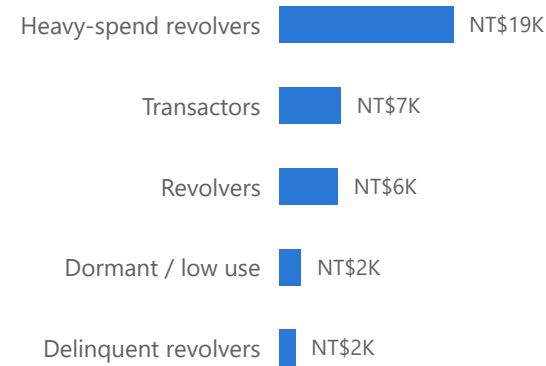
How many customers are in each segment?



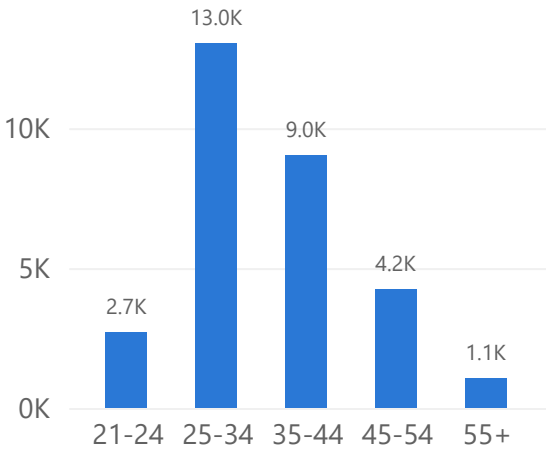
How do segments use the card?
Utilization vs share of bill paid (size = customers)



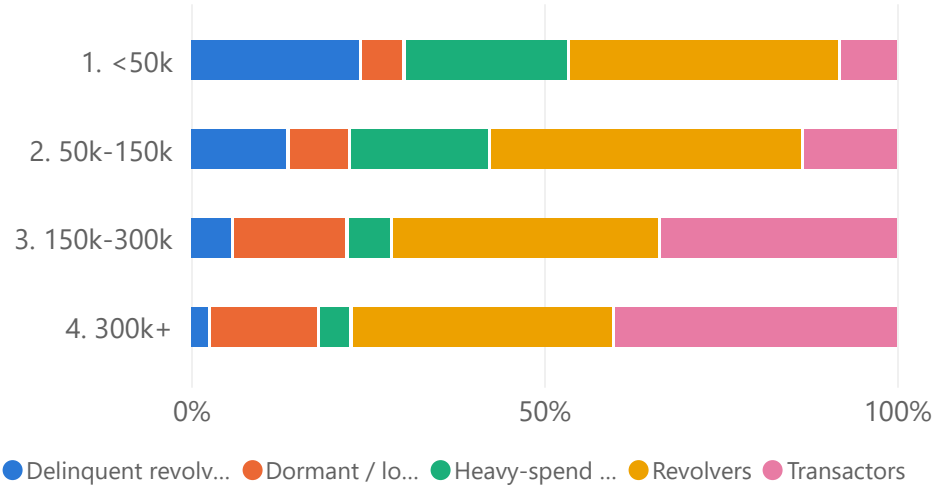
How much do they spend? Est. monthly purchases per customer



Demographics: customers by age band



Product usage: segment mix within each credit-limit tier



Segment summary

Segment	Customers	Utilization	Purchases / mo	Default rate	P(dormant)	Risk-adj. value / mo
Revolvers	12,073	53.0%	NT\$6,490	16.3%	0.3%	NT\$590
Heavy-spend revolvers	4,026	61.4%	NT\$19,210	20.3%	0.3%	NT\$284
Transactors	7,181	4.2%	NT\$6,827	14.4%	2.7%	-NT\$25
Delinquent revolvers	3,156	61.2%	NT\$1,794	62.8%	0.5%	-NT\$63
Dormant / low use	3,564	2.5%	NT\$2,425	23.4%	7.1%	-NT\$126
Total	30,000	37.3%	NT\$7,301	22.1%	1.4%	NT\$248

Risk analytics

PD = out-of-fold LightGBM probability of missing the Oct 2005 payment. Where is the credit risk and how much could it cost? (The data has no geography or card-product field.)

22.1%

Avg predicted PD

22.1%

Default rate (Oct, actual)

NT\$10.2M

Expected loss / mo (est.)

3,706

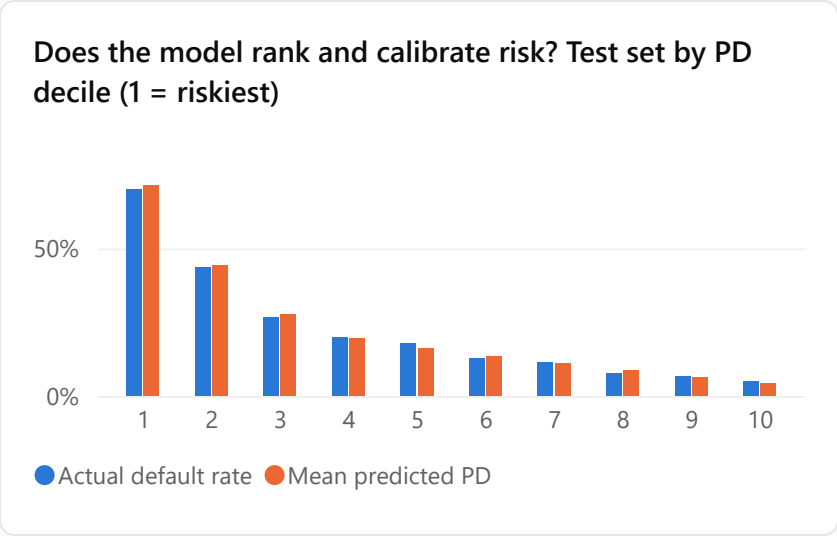
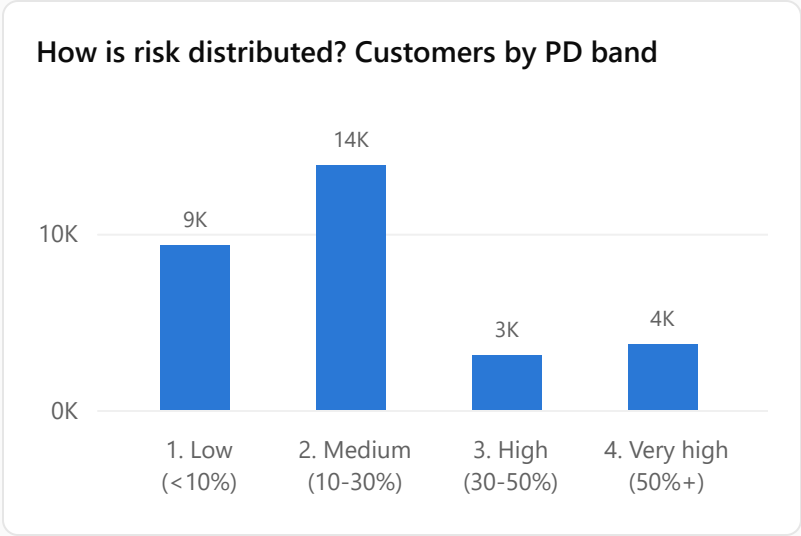
Customers with PD 50%+

0.54%

Expected loss / EAD

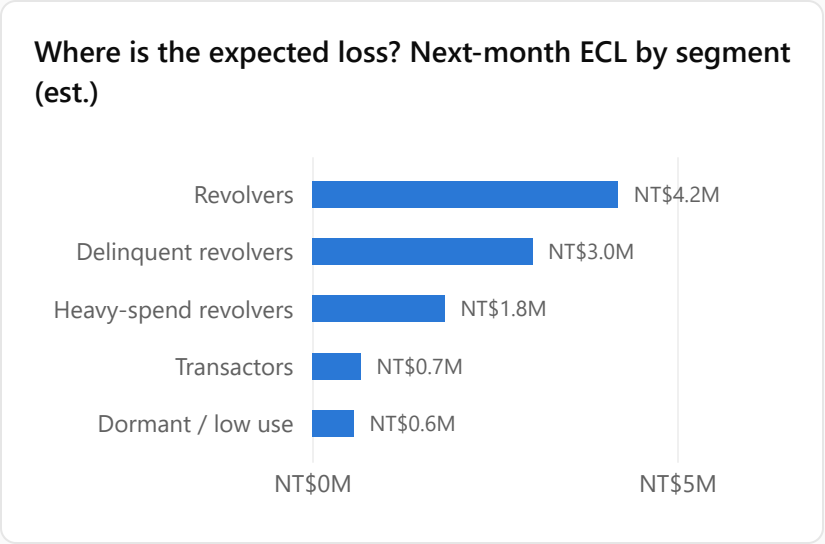
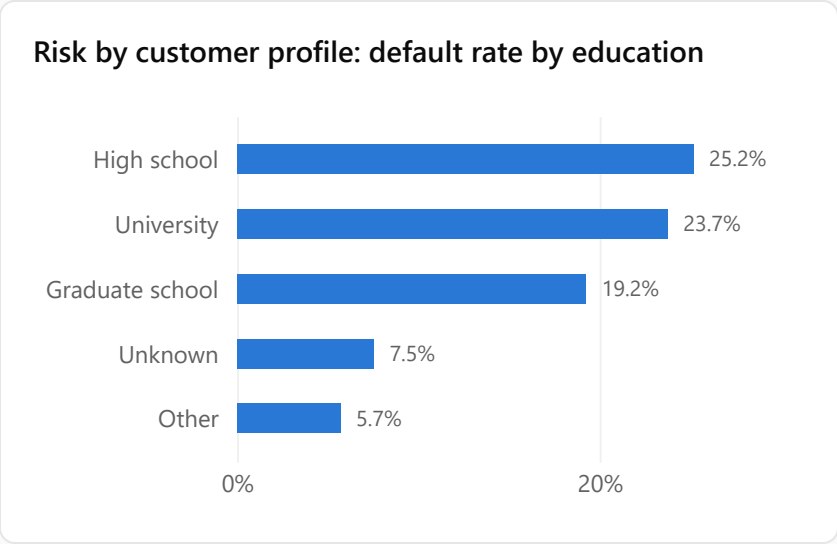
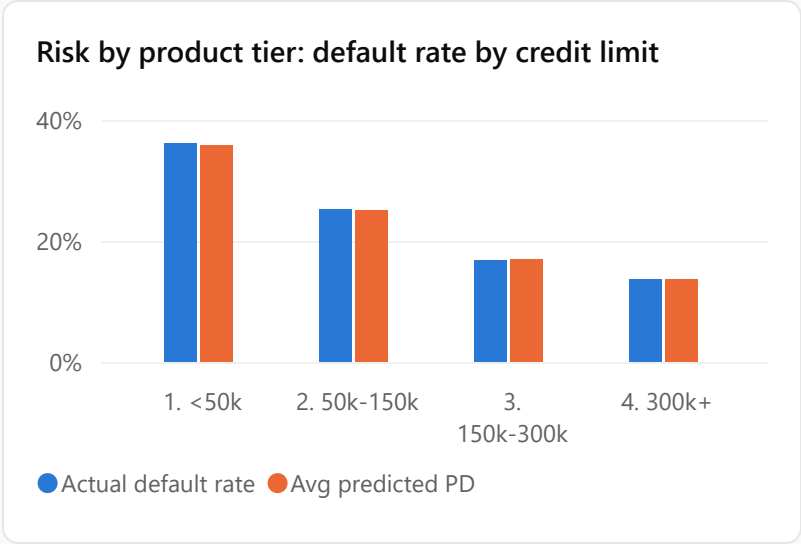
0.785

PD model test ROC-AUC



Roll rates: share moving to each status next month (Apr-Aug average)

from_bucket	0. Current	2. 60 DPD	3. 90+ DPD
0. Current	95%	5%	
2. 60 DPD	30%	64%	6%
3. 90+ DPD	10%	32%	58%



Retention decision support

Which customers should we prioritize, and why? Expected net benefit = P(dormant) x save rate x 12-month risk-adjusted value - contact cost. High-PD customers go to risk management instead.

Priority tier

All

Segment

All

28K

Active customers scored

57

Contact now

NT\$5,501

Net benefit of 'Contact now' (est.)

NT\$282.1K

Dormancy value at risk, 12m (est.)

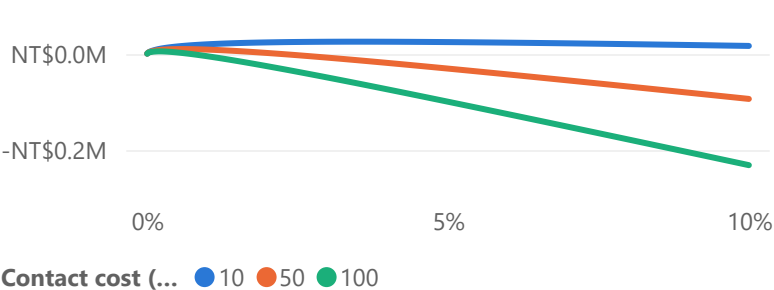
3,668

Excluded: PD 50%+

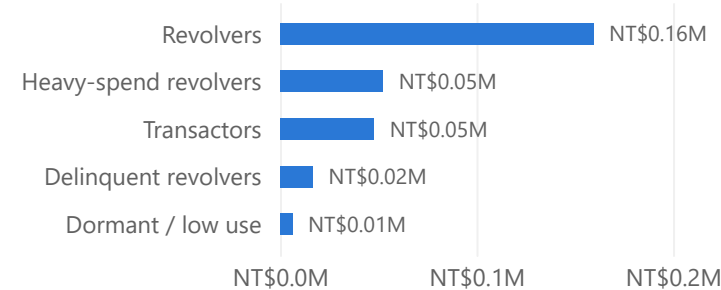
What happens to each active customer? Customers by priority tier



What if only 10% can be contacted? Cumulative expected net benefit by contact cost (NT\$)



Where is the dormancy value at risk? 12-month value at risk by segment (est.)



Decision list: who to contact first, and why (sorted by expected net benefit)

Customer	Segment	Value / month (est.)	P(dormant)	PD	Expected net benefit	Priority	Why
5297	Heavy-spend revolvers	NT\$3,026	9.8%	10.0%	NT\$609	1. Contact now	P(dormant) 9.8% PD 10% risk-adj. value
8501	Revolvers	NT\$3,970	6.7%	15.7%	NT\$535	1. Contact now	P(dormant) 6.7% PD 16% risk-adj. value
25950	Heavy-spend revolvers	NT\$1,806	12.3%	7.9%	NT\$433	1. Contact now	P(dormant) 12.3% PD 8% risk-adj. value 2%
20963	Revolvers	NT\$818	22.0%	27.7%	NT\$332	1. Contact now	P(dormant) 22.0% PD 28% risk-adj. valu
8323	Transactors	NT\$1,107	13.5%	7.8%	NT\$258	1. Contact now	P(dormant) 13.5% PD 8% risk-adj. value
29004	Revolvers	NT\$2,615	5.7%	9.1%	NT\$258	1. Contact now	P(dormant) 5.7% PD 9% risk-adj. value I
28042	Revolvers	NT\$1,433	10.1%	6.1%	NT\$248	1. Contact now	P(dormant) 10.1% PD 6% risk-adj. value

Assumptions

Save rate 20% (unknown - must be measured with a random holdout group).
Contact cost NT\$100; value horizon 12 months; budget scenario 10% of active customers.
Expected loss calibrated to a 8% annual loss rate on balances (sensitivity 4-12% in the docs).
The dormancy model predicts who is likely to go dormant, not who will respond to an offer.
At NT\$100 per contact only a few dozen customers are worth contacting; cheap digital nudges (about NT\$10) make sense for roughly 1,000.